

Wizarding World's Railway Ridership

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Executive Summary

In our **first draft** of the Project Proposal, we received feedback regarding fixing naming errors and suggestions for changes in entity relationships. After reviewing all feedback, we decided to:

- Remove the 1:M relationship between Routes and Passengers.
- Create a 1:M relationship between Trains and Passengers.
- Fixed entity and attribute naming errors for clarity.

Our **second draft** feedback mainly consisted of naming inconsistencies and data type suggestions. Changes made to the database and outline included:

- Added labels and comments for CRUD in the DML.SQL
- Fixed typos and inconsistencies between database and outline
- Revised attributes to be unique between tables (e.g. Stations “name” to “stationName”)
- Changed Passenger bloodStatus to ENUM (to hold W/M) wizard/muggle instead of VARCHAR(50) for simplicity

During the **normalization process**, we verified the database to 1NF and 2NF requirements, but did not continue to 3NF for this case. Adjustments made included:

- Remove the Train attribute, departureTime, and move it to Stations/Routes
- Create an intersection table Schedule to connect Stations and Routes in place of departureTime

For our **third draft**, we finally implemented our web app with structure to connect it to the database. Feedback suggested adding more dynamic features like editable text boxes in the table after the “edit” button is clicked. Changes included:

- Implemented working RESET features into the web app UI and a DELETE on the Passengers table to view the RESET.
- Created functions and views in PL.SQL callable as stored procedures in app.js

After receiving the feedback from step 4, we developed our **final draft**. Most feedback contained suggestions for UI and design improvements. Final improvements included:

- Adding CUD operations for the Schedules intersection table.
- Fixing bugs identified during Step 4 reviews to improve stability and functionality.

Overview

After Lord Voldemort was defeated, the number of witches and wizards attending Hogwarts has reached a new high. This has resulted in an unexpected uptick in the amount of ridership on the Hogwarts Express and other such lines throughout the Wizarding World. As such, the Ministry of Magic has decided to create a new database to better catalog trains and ridership.

Everyday, 20 trains transport upwards of 1000 passengers between 10 stations, catering to Hogwarts students and other commuters in the Wizarding World². Each passenger that registers at the station is registered as a transaction to keep track of ridership. The database aims to keep track of trains, train passengers, stations, and the routes available. A database driven website will allow for Hogwarts Express admin to swiftly view train connections, route availability, and passenger demographics to better serve commuters.

Database Outline

- **Trains:** Represents an individual magical train on the Wizarding Railway.
 - idTrain: INT, PK, unique, not NULL, auto_increment
 - trainName: VARCHAR(50), unique, not NULL
 - maxCapacity: INT, not NULL
 - idRoute: FK, INT, not NULL

Relationships:

- 1:M relationship between Trains and Passengers is implemented with idTrain as FK inside Passengers. One train can have many passengers.
- M:1 relationship between Trains and Routes is implemented with idRoute as a FK inside of Trains. One route can have many trains.

Constraints:

- Passengers assigned to one train cannot exceed the max capacity of the train.

- **Routes:** Represents a unique path between two magical stations.
 - idRoute: INT, PK, unique, not NULL, auto_increment
 - routeName: VARCHAR(50), unique, not NULL
 - duration: TIME, not Null
 - frequency: VARCHAR(50), not NULL

Relationships:

- 1:M relationship between Routes and Trains is implemented with idRoute as FK inside of Train. One route can have many trains.
- M:M relationship between Routes and Stations is implemented with both idRoute and idStation as FK inside the RouteStation intersection table. One route can have many stations and one station can have many routes.

Constraints:

- Duration cannot be 0

- **Passengers:** Represents a witch or wizard who can ride on a magical train.

- idPassenger: INT, PK, unique, not NULL, auto_increment
- firstName: VARCHAR(50), not NULL
- lastName: VARCHAR(50), not NULL
- house: VARCHAR(50)
- bloodStatus: ENUM
- idTrain: FK, INT, not NULL

Relationships:

- M:1 relationship between Passengers and Trains is implemented with idTrain as FK inside Passengers. One train can have many passengers.

- **Stations:** Represents a magical train station in the wizarding world

- idStation: INT, PK, unique, not NULL, auto_increment
- stationName: VARCHAR(50), unique, not NULL
- locationDescription: VARCHAR(500), not NULL
- muggleAccess: BOOLEAN, not Null

Relationships:

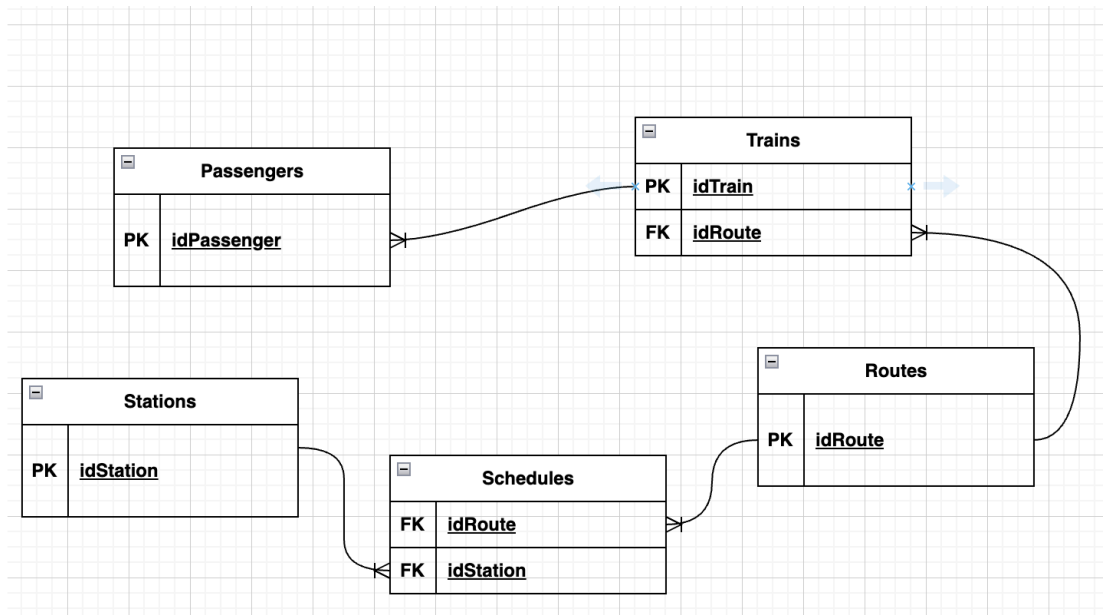
- M:M relationship between Stations and Routes is implemented with both idStation and idRoute as FK inside the RouteStation intersection table. One route can have many stations and one station can have many routes.

- **Schedules**

- idRoute: FK, INT, not NULL
- idStation: FK, INT, not NULL
- arrivalTime: DATETIME, not NULL
- departureTime: DATETIME, not NULL

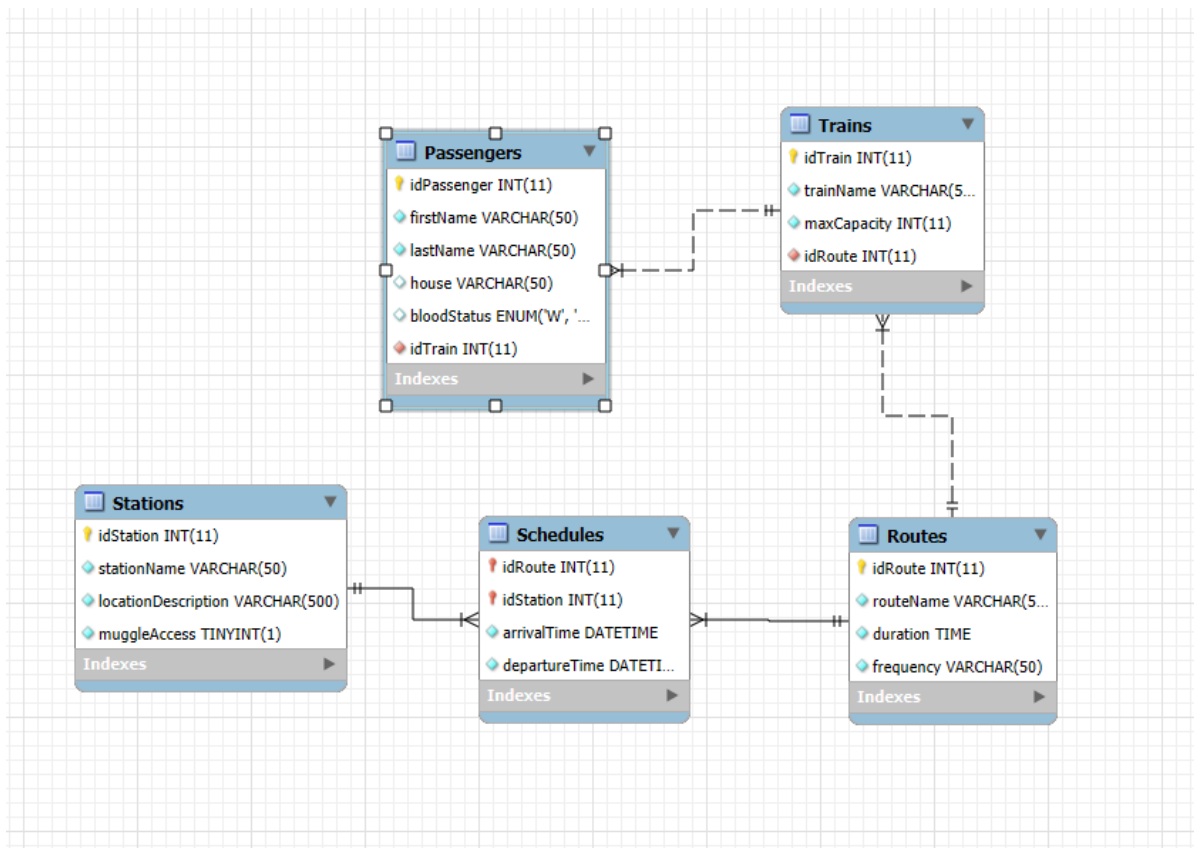
Relationships: intersection table connecting Stations and Routes.

Entity Relationship Diagram



Citation 4

Schema Diagram



Citation 3

Sample Data

Trains				
idTrain	name	maxCapacity	idRoute	
1	The Hogwarts Express	500	4	
2	The Silver Antler	300	3	
3	Emberlash Express	200	2	
4	Hearthlight Carriage	100	1	

Passengers					
idPassenger	firstName	lastName	house	bloodStatus	idTrain
1	Harry	Potter	Gryffindor	W	2
2	Hermione	Granger	Gryffindor	W	4
3	Draco	Malfoy	Slytherin	W	3
4	Dudley	Dursley		M	1

Stations				
idStation	name	locationDescription	muggleAccess	
1	Platform 9 3/4	located in the muggle world to access the wizarding world	1	
2	Antler Hollow	located in north wizarding village to take passengers to Hogwarts	0	
3	Emberlash Cross	located in a remote magical settlement	0	
4	Hearthlight Station	located in the heart of the city for magical commuters	0	

Routes				
idRoute	name	duration	duration	frequency

				y
1	The Hearthlight Circuit	30 minutes	00:30:00	12 daily
2	The Emberlash Loop	4 hours	04:00:00	5 daily
3	The Antler Passage	5 hours	05:00:00	3 daily
4	The Schoolbound Path	8 hours	08:00:00	4 yearly
Schedules				
idRoute	idStation	arrivalTime	departureTime	
1	4	2025-09-01 11:00:00	2025-09-01 11:02:00	
2	3	2025-07-24 8:30:00	2025-07-24 8:35:00	
3	2	2025-07-25 13:00:00	2025-07-25 13:02:00	
4	1	2025-07-24 17:00:00	2025-07-24 17:10:00	
4	2	2025-07-24 17:30:00	2025-07-24 17:35:00	

UI Screen Captures

Read Passengers

Passengers

idPassenger	firstName	lastName	house	bloodStatus	ridingTrain	
1	Harry	Potter	Gryffindor	W	The Silver Antler	<button>Delete</button>
2	Hermione	Granger	Gryffindor	W	Hearthlight Carriage	<button>Delete</button>
3	Draco	Malfoy	Slytherin	W	Emberlash Express	<button>Delete</button>
4	Dudley	Dursley		M	The Hogwarts Express	<button>Delete</button>

Read Trains

Trains

idTrain	trainName	maxCapacity	onRoute
1	The Hogwarts Express	500	The Schoolbound Path
2	The Silver Antler	300	The Antler Passage
3	Emberlash Express	200	The Emberlash Loop
4	Hearthlight Carriage	100	The Hearthlight Circuit

Read Routes

Routes

idRoute	routeName	duration	frequency
1	The Hearthlight Circuit	00:30:00	12 daily
2	The Emberlash Loop	04:00:00	5 daily
3	The Antler Passage	05:00:00	3 daily
4	The Schoolbound Path	08:00:00	4 yearly

Read Schedules

Schedules

Route	Station	Arrival Time	Departure Time	Delete	Edit
The Hearthlight Circuit	Hearthlight Station	Mon Sep 01 2025 11:00:00 GMT-0700 (Pacific Daylight Time)	Mon Sep 01 2025 11:02:00 GMT-0700 (Pacific Daylight Time)	<button>Delete</button>	<button>Edit</button>
The Emberlash Loop	Emberlash Cross	Fri Aug 15 2025 02:09:00 GMT-0700 (Pacific Daylight Time)	Sat Aug 16 2025 02:09:00 GMT-0700 (Pacific Daylight Time)	<button>Delete</button>	<button>Edit</button>
The Antler Passage	Antler Hollow	Fri Jul 25 2025 13:00:00 GMT-0700 (Pacific Daylight Time)	Fri Jul 25 2025 13:02:00 GMT-0700 (Pacific Daylight Time)	<button>Delete</button>	<button>Edit</button>
The Schoolbound Path	Platform 9 3/4	Thu Jul 24 2025 17:00:00 GMT-0700 (Pacific Daylight Time)	Thu Jul 24 2025 17:10:00 GMT-0700 (Pacific Daylight Time)	<button>Delete</button>	<button>Edit</button>
The Schoolbound Path	Antler Hollow	Thu Jul 24 2025 17:30:00 GMT-0700 (Pacific Daylight Time)	Thu Jul 24 2025 17:35:00 GMT-0700 (Pacific Daylight Time)	<button>Delete</button>	<button>Edit</button>
Select Route	Select Station	mm/dd/yyyy, --:-- --	mm/dd/yyyy, --:-- --	Add Schedule	

Read Stations

Stations

idStation	stationName	locationDescription	muggleAccess
1	Platform 9 3/4	located in the muggle world to access the wizarding world	1
2	Antler Hollow	located in north wizarding village to take passengers to Hogwarts	0
3	Emberlash Cross	located in a remote magical settlement	0
4	Hearthlight Station	located in the heart of the city for magical commuters	0

New Station

Create Station

Add New Station

Station Name:

Location Description:

Muggle Access:

No

Add Station

Create Schedule (Dropdown menu of Route/Station rows)

Route	Station	Arrival Time	Departure Time	Delete	Edit
The Hearthlight Circuit	Platform 9 3/4	Fri Aug 15 2025 02:18:00 GMT-0700 (Pacific Daylight Time)	Sat Aug 16 2025 02:18:00 GMT-0700 (Pacific Daylight Time)	Delete	Edt
The Emberlash Loop	Emberlash Cross	Fri Aug 15 2025 02:09:00 GMT-0700 (Pacific Daylight Time)	Sat Aug 16 2025 02:09:00 GMT-0700 (Pacific Daylight Time)	Delete	Edt
The Antler Passage	Antler Hollow	Fri Jul 25 2025 13:02:00 GMT-0700 (Pacific Daylight Time)	Sat Aug 16 2025 13:02:00 GMT-0700 (Pacific Daylight Time)	Delete	Edt
The Schoolbound Path	Platform 9 3/4	Thu Jul 24 2025 17:10:00 GMT-0700 (Pacific Daylight Time)	Fri Aug 15 2025 17:10:00 GMT-0700 (Pacific Daylight Time)	Delete	Edt
The Schoolbound Path	Antler Hollow	Thu Jul 24 2025 17:35:00 GMT-0700 (Pacific Daylight Time)	Fri Aug 15 2025 17:35:00 GMT-0700 (Pacific Daylight Time)	Delete	Edt

✓ Select Route

The Hearthlight Circuit

The Emberlash Loop

The Antler Passage

The Schoolbound Path

✓ Select Station

Platform 9 3/4

Antler Hollow

Emberlash Cross

Hearthlight Station

August 2025

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Clear

Today

mm/dd/yyyy, --:--:--

mm/dd/yyyy, --:--:--

Add Schedule

Reset database on Home page

Home page

Group 33

Logan Sailer and Lauren Gilane

After Lord Voldemort was defeated, the number of witches and wizards attending Hogwarts has reached a new high. This has resulted in an unexpected uptick in the amount of ridership on the Hogwarts Express and other such lines throughout the Wizarding World. As such, the Ministry of Magic has decided to create a new database to better catalog trains and ridership.

Everyday, 20 trains transport upwards of 1000 passengers between 10 stations, catering to Hogwarts students and other commuters in the Wizarding World 2. Each passenger that registers at the station is registered as a transaction to keep track of ridership. The database aims to keep track of trains, train passengers, stations, and the routes available. A database driven website will allow for Hogwarts Express admin to swiftly view train connections, route availability, and passenger demographics to better serve commuters.

Reset Database?

Success: Database reset successfully

Delete Schedule (clicking “Delete” button will remove the row)

Schedules

Route	Station	Arrival Time	Departure Time	Delete	Edit
The Hearthlight Circuit	Hearthlight Station	Mon Sep 01 2025 11:00:00 GMT-0700 (Pacific Daylight Time)	Mon Sep 01 2025 11:02:00 GMT-0700 (Pacific Daylight Time)	Delete	Edit

Delete Passengers (clicking “Delete” will remove the row)

Passengers

idPassenger	firstName	lastName	house	bloodStatus	ridingTrain	
1	Harry	Potter	Gryffindor	W	The Silver Antler	Delete
2	Hermione	Granger	Gryffindor	W	Hearthlight Carriage	Delete
3	Draco	Malfoy	Slytherin	W	Emberlash Express	Delete
4	Dudley	Dursley		M	The Hogwarts Express	Delete

Update Schedule (Update row pops up underneath existing row)

Route	Station	Arrival Time	Departure Time	Delete	Edit
The Hearthlight Circuit	Hearthlight Station	Mon Sep 01 2025 11:00:00 GMT-0700 (Pacific Daylight Time)	Mon Sep 01 2025 11:02:00 GMT-0700 (Pacific Daylight Time)	Delete	Edit
✓ The Hearthlight Circuit The Emberlash Loop	Platform 9 ¾	mm/dd/yyyy, --:-- --	mm/dd/yyyy, --:-- --	Save	

Citations

1. MySQL, "MySQL Workbench," <https://www.mysql.com/products/workbench/>. Accessed Jul 2, 2025
2. Rowling, J. K. Harry Potter. Scholastic, 1999-2007. Accessed Jul 2, 2025
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<https://dbdiagram.io>
4. *draw.io*. JGraph Ltd., <https://www.draw.io>. Accessed Jul 31, 2025